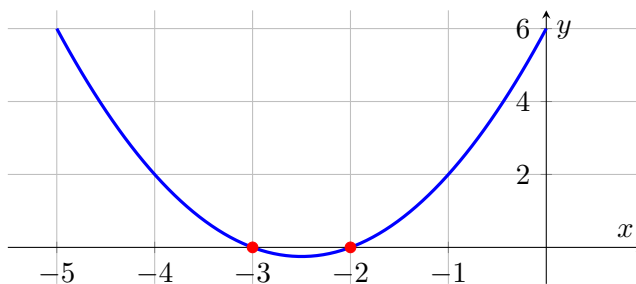


Vocabulary**Zero Product Property:** if $A \cdot B = 0$, then $A = 0$ or $B = 0$.**Root (solution):** an x that makes the equation true; on a graph it is an x-intercept.**We do together.** Solve $x^2 + 5x + 6 = 0$.**Step 1.** One side already = 0. Good.**Step 2.** Factor: $x^2 + 5x + 6 = (x + 2)(x + 3)$ **Step 3.** Set each factor = 0: $x + 2 = 0$ or $x + 3 = 0$ **Step 4.** Solve: $x = -2$ or $x = -3$ **See it:** the roots are where the parabola crosses the x -axis.**You Try.** Solve each. Show every step, then name the roots.

a) $x^2 - x - 12 = 0$

$(x - 4)(x + 3) = 0 \Rightarrow x = 4, -3$

b) $x^2 - 9 = 0$

$(x - 3)(x + 3) = 0 \Rightarrow x = 3, -3$

Summary (fill in): To solve a quadratic by factoring: make one side = 0, factor the other side, then use the Zero Product Property and solve each factor.